

Space-Time Π Identity and the Integrated Quadratic Boundary Oscillation (Route δ , Plan 2, Building Block **SPACE****TIME****IDENTITY**)

Abstract

We prove the space-time radial-moment identity

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu(\rho) = (n-1) \int_{\tilde{\Omega}} \sigma(x) dx,$$

together with the integrated quadratic Fusco–Julin boundary oscillation

$$\int_{\rho_*}^{\rho_\delta} \int_{\partial^* E_\rho} |\nu_{E_\rho} - e_{r, \bar{z}_\rho}|^2 d\mathcal{H}^{n-1} d\mu(\rho) \leq C(n, R, \rho_*) \delta_T(\Omega), \quad (\text{B}^2\text{-int})$$

and the integrated radial trace

$$\int_{\rho_*}^{\rho_\delta} \left(\int_{\partial^* E_\rho} |H_{\bar{z}_\rho, \rho} - 1| d\mathcal{H}^{n-1} \right)^2 d\mu(\rho) \leq C(n, R, \rho_*) \delta_T(\Omega), \quad (\text{T}_1^2\text{-int})$$

conditional only on the H^1 -in- ρ centroid bound (F) of [3]. The proof of $(\text{T}_1^2\text{-int})$ uses a *slicing-then-squaring* decomposition $T_1 = T_1^{\text{rad, slic}} + T_1^{\text{tan}}$, in which the radial piece is bounded *linearly* (not at the FMP rate $\sqrt{\delta}$) via the good-ray equality of slicewise volumes and the bad-ray multiplicity bound of [1]; the squared volume Fraenkel $|E_\rho \Delta B_\rho|^2$ is controlled at rate δ via the new pointwise slice-rearrangement inequality $|E_\rho \Delta B_\rho|^2 \leq C \Pi(E_\rho, \bar{z}_\rho)$. The per- ρ radial L^1 trace (R) is genuinely *not* used.

1 Notation and statement of the targets

1.1 Standing hypotheses

Fix $n \geq 2$, $R \geq 2$, $\rho_* \in (0, 1)$. Let $\Omega \subset B_R$ open with $|\Omega| = \omega_n$, $\delta := \delta_T(\Omega) \leq \delta_0(n, R, \rho_*)$. Let $u = u_\Omega$ be the variational torsion function, $t : [\rho_*, 1] \rightarrow (0, \|u\|_\infty)$ the volume-radius parametrisation, $E_\rho := \{u > t(\rho)\}$, $t_B(\rho) := (1 - \rho^2)/(2n)$, $\psi(\rho) := -t'(\rho) - (-t'_B(\rho)) \geq 0$, $\rho_\delta := 1 - \kappa\sqrt{\delta}$, $\mu(d\rho) := (-t'(\rho))d\rho$.

For each ρ , $\bar{z}_\rho = \bar{z}_\rho$ is the centroid of E_ρ ([3] Theorem F). By [1] Lemma 2.2, $B(\bar{z}_\rho, \rho/4) \subset E_\rho \subset B(\bar{z}_\rho, 3\rho)$, in particular $\rho_*/2 \leq |x - \bar{z}_\rho| \leq 2R$ on $\partial^* E_\rho$.

The L^2 divergence identity of [1] (5.3):

$$\int_{\partial^* E_\rho} |\nu_{E_\rho} - e_{r, \bar{z}_\rho}|^2 d\mathcal{H}^{n-1} = 2(P(E_\rho) - P(B_\rho)) + 2\Pi(E_\rho, \bar{z}_\rho), \quad (1)$$

where

$$\Pi(E_\rho, \bar{z}_\rho) := (n-1) \int_{B_\rho(\bar{z}_\rho) \setminus E_\rho} |x - \bar{z}_\rho|^{-1} dx - (n-1) \int_{E_\rho \setminus B_\rho(\bar{z}_\rho)} |x - \bar{z}_\rho|^{-1} dx \geq 0. \quad (2)$$

1.2 The three target theorems

Theorem 1.1 (Space-time Π identity, (G1)). $\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu(\rho) = (n-1) \int_{\tilde{\Omega}} \sigma(x) dx$ for a signed Fubini density σ with $|\sigma(x)| \leq (4/n\rho_*) \mathcal{L}^1(J_x)$, where $J_x := \{\rho : x \in B_\rho(\bar{z}_\rho) \Delta E_\rho\}$.

Theorem 1.2 (Integrated (B²-int), (G3)). Conditional on (F),

$$\int_{\rho_*}^{\rho_\delta} \int_{\partial^* E_\rho} |\nu_{E_\rho} - e_{r, \bar{z}_\rho}|^2 d\mathcal{H}^{n-1} d\mu(\rho) \leq C_{B^2}(n, R, \rho_*) \delta_T(\Omega).$$

Theorem 1.3 (Integrated radial trace, (T₁²-int)). Conditional on (F),

$$\int_{\rho_*}^{\rho_\delta} \left(\int_{\partial^* E_\rho} |H_{\bar{z}_\rho, \rho} - 1| d\mathcal{H}^{n-1} \right)^2 d\mu(\rho) \leq C_{\text{radial}}(n, R, \rho_*) \delta_T(\Omega).$$

2 Space-time identity (G1)

By (2), $\Pi(E_\rho, \bar{z}_\rho) = (n-1) \int_{\mathbb{R}^n} (\mathbf{1}_{B_\rho(\bar{z}_\rho)} - \mathbf{1}_{E_\rho}) |x - \bar{z}_\rho|^{-1} dx$. By joint Borel measurability and Fubini,

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu(\rho) = (n-1) \int_{\mathbb{R}^n} \sigma(x) dx, \quad (3)$$

where $\sigma(x) := \int_{\rho_*}^{\rho_\delta} (\mathbf{1}_{B_\rho(\bar{z}_\rho)} - \mathbf{1}_{E_\rho}) / |x - \bar{z}_\rho| d\mu(\rho)$. By CL containment, the integrand vanishes outside $\tilde{\Omega} := \Omega \cup B_{3R}$ and $|x - \bar{z}_\rho| \geq \rho_*/2$ on its support; combined with $-t' \leq 1/n$, $|\sigma(x)| \leq (4/n\rho_*) \mathcal{L}^1(J_x)$. $\square$

3 Profile-gap conversion with centre drift (G2)

By Talenti / Nicola–Tilli profile gap [1] Lemma 8.1, $0 \leq t_B(\rho) - t(\rho) \leq 2\delta/(\omega_n \rho_*^n)$. With $\tau_0 := \rho_*/(2n)$ and on the good profile set $\{-t' \geq \tau_0\}$, $|\rho_E(x) - \rho_B^*(x)| \leq C_T \delta$ by MVT.

For moving centre, use $\varphi_x(\rho) := |x - \bar{z}_\rho| - \rho$; on $\{|\bar{z}'_\rho| \leq 1/2\}$, $\varphi'_x \leq -1/2$. The good/bad decomposition via Chebyshev on $\{|\bar{z}'_\rho| > 1/2\}$ controlled by (F) gives:

$$\mathcal{L}^1(J_x) \leq C_T \delta + \int |\bar{z}'_\rho| \mathbf{1}_{A_x} d\rho + 4 \int |\bar{z}'_\rho|^2 d\rho. \quad (4)$$

4 Assembly of (B²-int)

By (1) integrated,

$$\int |\nu - e_{r, \bar{z}_\rho}|^2 dH d\mu = 2 \int (P - P_\rho) d\mu + 2 \int \Pi(E_\rho, \bar{z}_\rho) d\mu.$$

The first piece is $\leq C\delta$ via Agent 1's deficit identity. The second piece is bounded via (G1)+(G2)+(F)+Cauchy–Schwarz:

$$\int \Pi(E_\rho, \bar{z}_\rho) d\mu \leq \frac{4(n-1)}{n\rho_*} \int_{\tilde{\Omega}} \mathcal{L}^1(J_x) dx \leq C_2(n, R, \rho_*, C_F) \delta.$$

Both pieces $O(\delta)$ gives (B²-int). $\square$

5 The integrated radial trace (T_1^2 -int) — slicing-then-squaring

Write $T_1 = T_1(E_\rho, \bar{z}_\rho) := \int_{\partial^* E_\rho} |x - \bar{z}_\rho|/\rho - 1| d\mathcal{H}^{n-1}$ and decompose

$$T_1 = T_1^{\text{rad,slic}} + T_1^{\text{tan}},$$

with

$$\begin{aligned} T_1^{\text{rad,slic}} &:= \int_{\partial^* E_\rho} |x - \bar{z}_\rho|/\rho - 1| \cdot |e_{r,\bar{z}_\rho} \cdot \nu_{E_\rho}| d\mathcal{H}^{n-1}, \\ T_1^{\text{tan}} &:= \int_{\partial^* E_\rho} |x - \bar{z}_\rho|/\rho - 1| \cdot (1 - |e_{r,\bar{z}_\rho} \cdot \nu_{E_\rho}|) d\mathcal{H}^{n-1}. \end{aligned}$$

By the slicing identity [5, Thm 18.11] applied with $g(r) := |r/\rho - 1|$:

$$T_1^{\text{rad,slic}} = \int_{S^{n-1}} \sum_{j=1}^{N(\theta)} |r_{\theta,j}/\rho - 1| r_{\theta,j}^{n-1} d\theta. \quad (5)$$

5.1 Pointwise bound for $T_1^{\text{rad,slic}}$

Lemma 5.1. $T_1^{\text{rad,slic}} \leq C_4(n, R, \rho_*) [|E_\rho \Delta B_\rho(\bar{z}_\rho)| + (P(E_\rho) - P(B_\rho))]$.

Proof. Good rays. On a good ray ($N = 1$), the slicewise 1D symmetric-difference equals:

$$\int_0^\infty |\mathbf{1}_{E_{\rho,\theta}} - \mathbf{1}_{(0,\rho)}| r^{n-1} dr = \frac{1}{n} |r_{\theta,1}^n - \rho^n|. \quad (6)$$

By MVT, $|r_{\theta,1}/\rho - 1| \cdot r_{\theta,1}^{n-1} \leq \beta_n |r_{\theta,1}^n - \rho^n|$ with $\beta_n = \beta_n(R, \rho_*)$. Integrating and using spherical Fubini:

$$\int_{\{N=1\}} |r_{\theta,1}/\rho - 1| r_{\theta,1}^{n-1} d\theta \leq n\beta_n |E_\rho \Delta B_\rho(\bar{z}_\rho)|.$$

Bad rays. $|r_{\theta,j}/\rho - 1| r_{\theta,j}^{n-1} \leq (2R)^{n-1}$, summed gives $\leq (2R)^{n-1}(1+k(\theta))$. By the elementary bad-ray multiplicity bound $\int k(\theta) d\theta \leq (2/\rho_*)^{n-1}(P(E_\rho) - P(B_\rho))$, which follows from the slicing identity (5), the inner-ball containment, and the isoperimetric inequality (each extra crossing contributes $\geq (\rho_*/2)^{n-1}$ to $P(E_\rho)$ beyond the spherical baseline $P(B_\rho)$),

Combining gives the claim. $\square$

5.2 Pointwise bound for T_1^{tan}

Lemma 5.2. $T_1^{\text{tan}} \leq (P(E_\rho) - P(B_\rho)) + \Pi(E_\rho, \bar{z}_\rho)$.

Proof. By CL, $|x - \bar{z}_\rho|/\rho - 1| \leq 1$, so $T_1^{\text{tan}} \leq \int (1 - |e_{r,\bar{z}_\rho} \cdot \nu|) dH \leq \int (1 - e_{r,\bar{z}_\rho} \cdot \nu) dH = (P - P_\rho) + \Pi(E_\rho, \bar{z}_\rho)$ by [1] (5.3). $\square$

5.3 Squaring and integrating

By Lemmas 5.1, 5.2 and $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$:

$$T_1^2 \leq 3C_5^2 [|E_\rho \Delta B_\rho|^2 + (P - P_\rho)^2 + \Pi(E_\rho, \bar{z}_\rho)^2].$$

Integrating against $d\mu$ and bounding each piece by $C\delta$:

- $\int |E_\rho \Delta B_\rho|^2 d\mu \leq C_8 \int \Pi(E_\rho, \bar{z}_\rho) d\mu \leq C\delta$ via the slice-rearrangement (§6) + (G3).
- $\int (P - P_\rho)^2 d\mu \leq 2P(B_\rho) \int (P - P_\rho) d\mu \leq C\delta$ via Agent 1.

- $\int \Pi(E_\rho, \bar{z}_\rho)^2 d\mu \leq \Pi_{\max} \int \Pi(E_\rho, \bar{z}_\rho) d\mu \leq C\delta$ since Π is bounded by CL.

Hence $\int T_1^2 d\mu \leq C_{T_1^2} \delta$.

Combining with (G3) gives $(T_1^2\text{-int})$ since $|H - 1| \leq \|x - \bar{z}_\rho\|/\rho - 1 + |e_{r, \bar{z}_\rho} \cdot \nu - 1|$ and the second term is controlled by $|e_{r, \bar{z}_\rho} - \nu| + \text{Cauchy-Schwarz} + (\text{G3})$. $\square$

6 The slice-rearrangement inequality

Lemma 6.1. $|E_\rho \Delta B_\rho(\bar{z}_\rho)|^2 \leq C_8(n, R, \rho_*) \Pi(E_\rho, \bar{z}_\rho)$ pointwise in the smallness regime.

Proof. Step 1: $\Pi(E_\rho, \bar{z}_\rho) \geq cM_1$. Let $M_1 := \int_{E_\rho \Delta B_\rho} |\rho - |x - \bar{z}_\rho|| dx$. By rewriting (2),

$$\Pi(E_\rho, \bar{z}_\rho) = (n-1) \int_{E_\rho \Delta B_\rho} \frac{|\rho - |x - \bar{z}_\rho||}{\rho |x - \bar{z}_\rho|} dx \geq \frac{n-1}{2R\rho} M_1.$$

Step 2: $|E_\rho \Delta B_\rho|^2 \leq C M_1$. By polar coordinates and a slicewise Cauchy-Schwarz on $[\rho_*/2, 2R]$, $\sigma(\theta)^2 \leq 4R m_1(\theta)$ with $\sigma(\theta) = \mathcal{L}^1(E_{\rho, \theta} \Delta (0, \rho))$, $m_1(\theta) = \int |r - \rho| \mathbf{1}_{E_{\rho, \theta} \Delta (0, \rho)} dr$. Cauchy-Schwarz on S^{n-1} then gives $(\int \sigma d\theta)^2 \leq 4R |S^{n-1}| \int m_1 d\theta$. Boosting slice integrals to global integrals via the r^{n-1} weight bounds: $|E_\rho \Delta B_\rho|^2 \leq C_9 M_1$.

Combining Steps 1–2 gives the claim. $\square$

7 Discussion: why T_1 -route closes where T_2 -route stalls

Remark 7.1. The T_2 -route would extract on good rays $(r_{\theta,1}/\rho - 1)^2 r_{\theta,1}^{n-1} \leq C |r_{\theta,1}^{n-1} - \rho^{n-1}|$ (a square on LHS, unsigned mismatch on RHS), then need $\int |r_{\theta,1}^{n-1} - \rho^{n-1}| d\theta \leq C \sqrt{\delta_\rho}$ (FMP rate). Integrating against $d\mu$ leaks at $\sqrt{\delta}$.

The T_1 -route uses linear good-ray equality (6) and the slice-rearrangement to get $|E_\rho \Delta B_\rho|^2 \leq C\Pi$, which integrates to $C\delta$ via (G3). Squaring *before* integrating avoids the FMP bottleneck.

8 Conditionality

Inputs used (all unconditional given (F)): CL containment; slicing identity (Maggi); bad-ray multiplicity bound (elementary, see proof of Lemma 5.1); slicewise good-ray equality; spherical Fubini; divergence identity (5.3); Talenti profile gap; slice-rearrangement Lemma 6.1; (F) of [3].

Inputs explicitly not used: the per- ρ pointwise radial-trace inputs (1.1), (1.2)=(R), (2.4); pointwise (B²); Agent 3 (7.1); any selection, graph entry, Schauder, Fuglede.

References

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